

कुटुंब

KUTUMB

Ghar se door, dil ke paas.

Away from home, close to the heart.

AI-powered family health companion
Voice-first · 100+ languages · Privacy-first

Amrit · Nishita

HopeWorks Hackathon · April 2026

1.4B

Indians to reach first

7,100

Languages to serve globally

₹0 / \$0

Health data stored on our servers

MEET THE TEAM

Two builders. One mission. A billion families — and a world that needs better health tools.

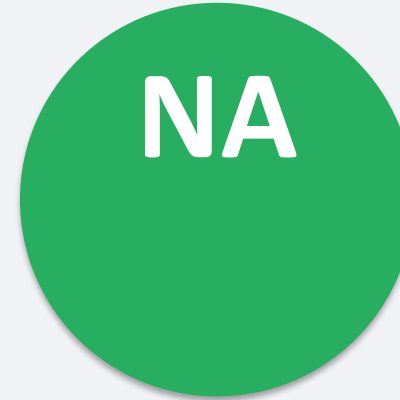


Amrit Lahari

Full-Stack Developer & AI Engineer

B.Tech 3rd Year · BITS Pilani

*FastAPI · Anthropic Claude SDK · Firebase
React Native · AES-256-GCM · System Architecture*



Nishita Agarwal

Product Designer & Frontend Engineer

B.Tech 3rd Year · BITS Pilani

*UI/UX Design · React Native · Voice UX
Multilingual UX Research · Data Viz · Accessibility*

"How many of us call home every day,
ask 'how are you?' and get 'I'm fine'

— even when it's not true."

THE GLOBAL PROBLEM

Healthcare systems are reactive, not proactive.
Families engage doctors only during crises.
Doctors receive zero longitudinal data.

India's doctor-patient ratio: 1:1456
— nearly 50% worse than WHO minimum.

4.5B

People lack essential health access (WHO 2023)

68%

Of global disease burden is preventable

\$8.3T

Annual economic loss from poor health outcomes

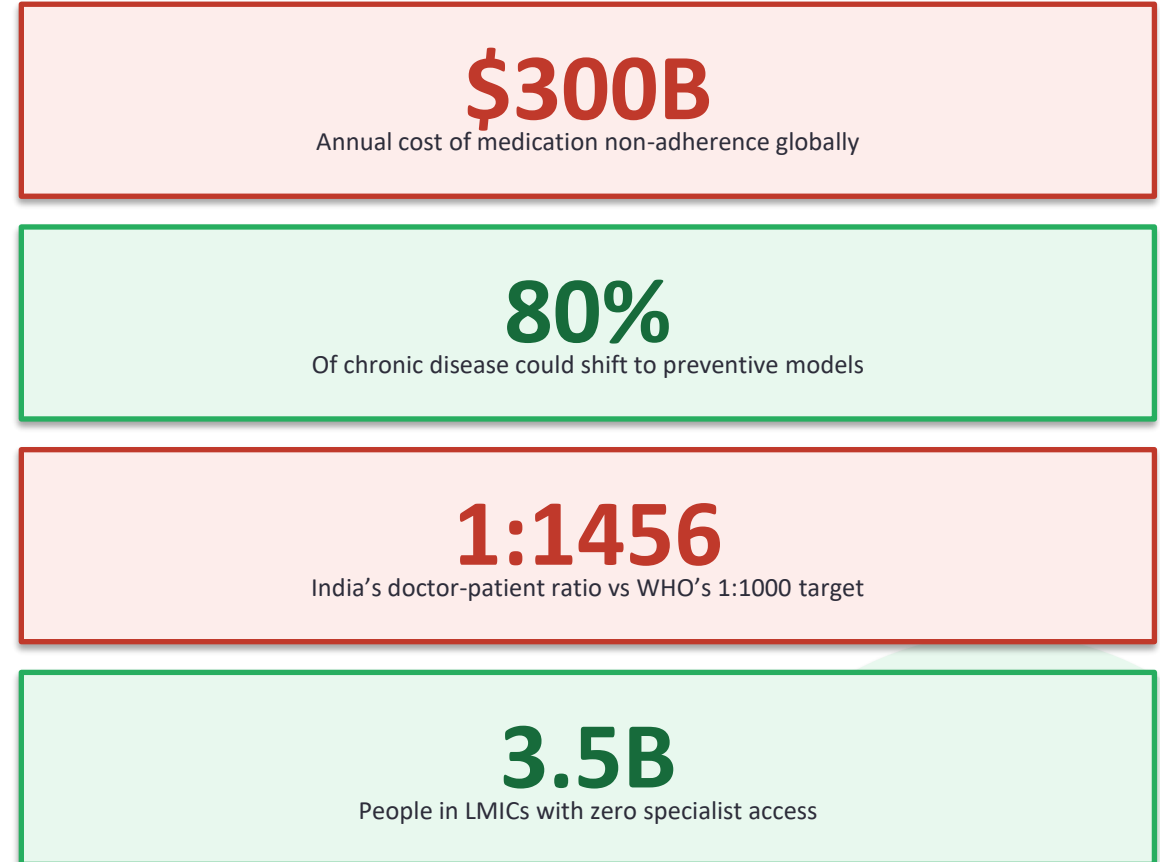
HOW THE PROBLEM SHOWS UP

Manifestation for users · Impact on the system

HOW IT MANIFESTS

- 01 Silent deterioration — ‘I’m fine’ hides months of worsening symptoms doctors never see
- 02 No longitudinal data at consultations. Doctors start fresh every single visit
- 03 Language exclusion: 40%+ of patients cannot describe symptoms in the healthcare system’s language
- 04 Medicine non-adherence costs \$300B/year (US alone) — 125,000 preventable deaths annually
- 05 Rural invisibility: 3.5B people in areas with zero specialist access

HOW IT IMPACTS



KUTUMB — THE SOLUTION

A Digital Health Custodian — we hear you when you don't mean your "I'm fine"s

What We Build

A voice-first AI family health companion.

- Listens in any dialect or language
- Monitors symptoms, vitals & medicines daily
- Alerts family in real time, anywhere on Earth
- Generates doctor-ready clinical summaries
- Triggers emergency SOS automatically
- Works offline in villages with no internet
- Health data stays on your phone only

1

Voice-First NLP

100+ Indian dialects
+ global language expansion

3

Genetic Risk Engine

Hereditary disease logic
auto-adjusts per member

5

Smart Medicine Cabinet

OCR scan · Interaction check
Dose reminders + refills

2

Family Dashboard

Real-time health view
of every family member

4

One-Tap SOS

Family + hospital + ambulance
alerted in under 60s

6

Doctor Referral PDF

Auto-generated longitudinal
health summary for visits

WHO WE SERVE

Three real humans. One platform.

R

Ramprasad Sharma

Retired teacher, 68 · Jaipur · Diabetic

Diabetic

Hypertensive

Tech-hesitant

Children abroad

NEED

Medicine reminders in Marwari,
reassure his children

PAIN

Says 'I'm fine' every call.
Forgets 3 medicines weekly.

GAIN

Medicine adherence +70%.
SOS to children in <60s.

P

Priya Mehta

Software engineer, 32 · Bengaluru / Chicago

Global caregiver

Time-starved

Data-driven

Helpless remotely

NEED

Real-time window into parents'
health from 8,000 miles

PAIN

Can't attend doctor visits.
No hereditary risk insight.

GAIN

Live family dashboard.
3h/week saved on remote care.

S

Savitribai Patil

ASHA worker, 38 · Vidarbha · Serves 200+ families

200+ families

No specialist

Offline-critical

Trusted locally

NEED

Clinical support in Marathi,
offline, for rural monitoring

PAIN

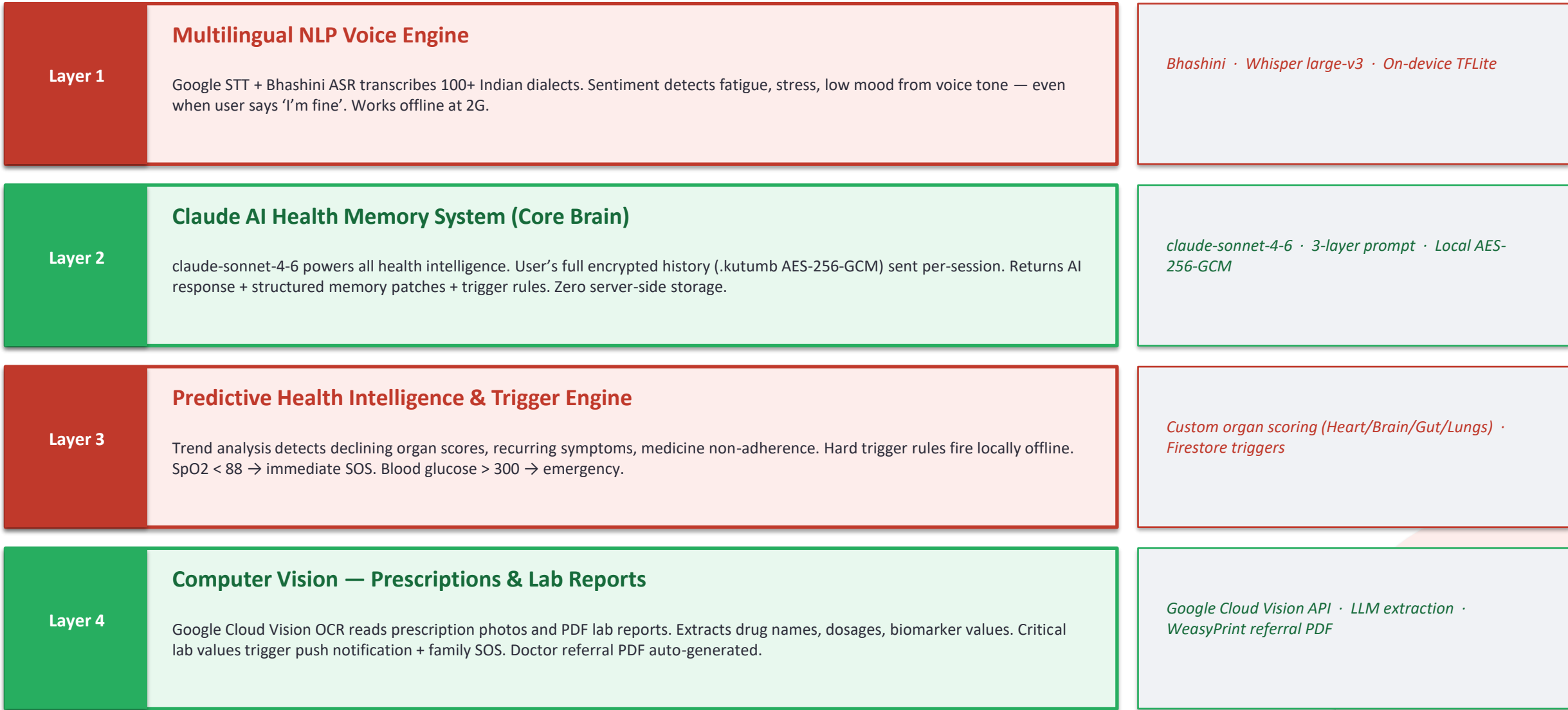
Nearest doctor 30 km away.
Paper records lost.

GAIN

200 families on screen.
Wrong referrals cut 50%.

THE AI BRAIN — 4-LAYER INTELLIGENCE STACK

Every layer has one job. Together they save lives.



LANGUAGE SCALABILITY — 7,100 LANGUAGES

Health advice that speaks your language, not just the hospital's.

7,100

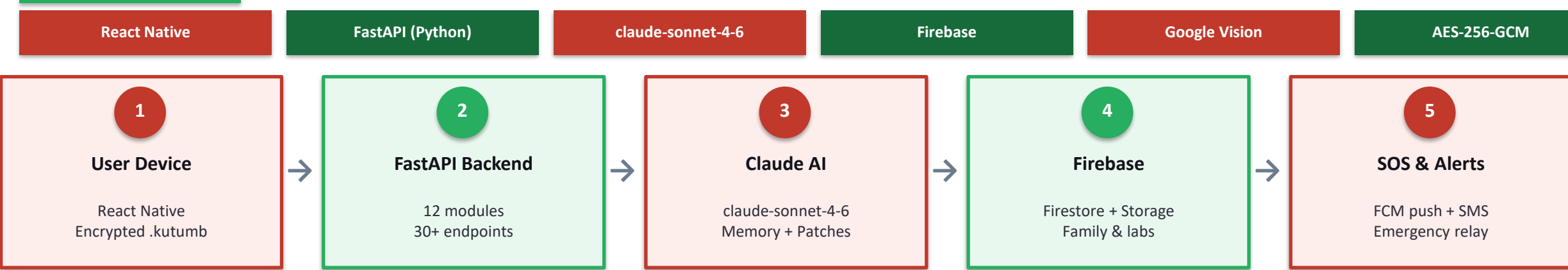
languages

“The same AI that asks ‘Did you take your BP medicine?’ in Marwari will ask the same question in Swahili, Hausa, Tagalog, and Haitian Creole—trained on the same architecture, deployed on the same app.”

TIER 1 Day 1	11 langs · 1.2B speakers Hindi · Marathi · Bengali · Tamil · Telugu · Kannada · Gujarati · Punjabi · Malayalam · Odia · English
TIER 2 3–6 months	10 langs · 800M+ speakers Swahili · Hausa · Amharic · Yoruba · Tagalog · Bahasa · Vietnamese · Burmese · Nepali · Sinhala
TIER 3 12–24 months	14 lang families · 3B+ speakers Spanish · Portuguese · Arabic · French (Africa) · Mandarin · Urdu · Pashto · Dari · Haitian Creole + dialects
TIER 4 36 months+	Full global coverage All 7,100 UNESCO-recognised languages via low-resource NLP, code-switching models, federated on-device fine-tuning

SYSTEM ARCHITECTURE

Privacy-first by design — health data never leaves the device



[LOCK] LOCAL HEALTH MEMORY: .kutumb file (AES-256-GCM) lives ONLY on user's device. Sent per-session over TLS 1.3. Purged from server memory immediately. DPDP · HIPAA · GDPR compliant by architecture.

✓ DPDP 2023 (India)

✓ HIPAA (United States)

✓ GDPR (European Union)

✓ Offline-First (2G)

12 BACKEND MODULES:

Auth	Users	Family	Check-ins	Medicines	Health Scores
Emergency SOS	Referrals PDF	Wearable Sync	Lab OCR	AI Insights	AI Session

TECHNICAL REQUEST — FACILITATED DATASETS

What we have · What we need · How it unlocks global reach

WHAT WE HAVE (Open Source, Already Live)



Bhashini Open API

Hindi / Marathi / Bengali STT live in prototype



SNOMED CT (free tier)

Clinical terminology, symptom tagging



RxNorm (NLM open)

Drug interaction baseline, name normalisation



ICMR OGD Portal

40 district health statistics for score calibration



Whisper open weights

Foundation model for all non-Bhashini languages

WHAT WE NEED (Facilitated / AI4India)



Bhashini ASR Training Corpus

500h/language · 12 languages · Speaker-level granularity (age, region, gender)



Anonymised Longitudinal Health Records

5 years · 100K+ patients · Urban + rural split · ICD-10 diagnosis codes



ICMR Lab Report Template Library

50+ formats · 10,000+ anonymised samples · Lab name, biomarker, value, reference range



Low-Resource Language Audio (LMICs)

100h/language: Swahili, Hausa, Tagalog, Bahasa · Health-domain sentences preferred

ENVIRONMENT & SECURITY REQUIREMENTS

Access methods · Compliance · Privacy by architecture

ACCESS METHOD & DATA DELIVERY



Secure API (Preferred)

HTTPS REST with OAuth2 / API-key. Data streamed, not cached. All responses via encrypted channel.



AIKosh Sandbox

Health records processed entirely inside the AIKosh secure sandbox. No data extraction. Results-only output.



Encrypted Download

Static corpora as AES-256 ZIP to isolated research environment. Destruction certification on completion.

COMPLIANCE ARCHITECTURE

✓ DPDP Act 2023 (India)

No health data stored server-side between sessions. Consent gate at onboarding. Right to erasure: delete account = wipe local .kutumb file.

✓ HIPAA (United States)

Zero PHI stored server-side. TLS 1.3 for all data in transit. Zero logging of health content. Business Associate Agreement available for US.

✓ GDPR (European Union)

Explicit opt-in consent. Data minimisation: only sha256 user hash sent to server. Right to portability (export .kutumb). Right to erasure.

BUSINESS MODEL, PARTNERSHIPS & RETENTION

Three tiers. Recurring revenue. Global partnership runway.

FREE ₹0 / \$0	PLUS ₹149 / \$2.99 mo	CARE PRO ₹499 / \$7.99 mo
✓ Daily voice check-in ✓ Family dashboard (3 members) ✓ Basic medicine reminder ✓ One-tap SOS ✓ Kutumb AI (5/day)	✓ Unlimited family members ✓ Full AI health memory ✓ Doctor referral PDFs ✓ Lab report OCR ✓ Wearable sync ✓ Priority ambulance SOS	✓ ASHA/CHW multi-family view ✓ Clinical decision support ✓ Govt. scheme integration ✓ Telemedicine booking ✓ White-label (NHS · CGHS · Medicaid)

RETENTION & PARTNERSHIPS



Pharmeasy / 1mg

Medicine refill + streak coupons



Thyrocare / Redcliffe

Lab booking from referral PDF



WHO / UNICEF

UHC 2030 alignment, LMIC rollout



NHS · CGHS · Medicaid

White-label government deployments

CHALLENGES & RISKS — WITH MITIGATIONS

We've thought hard about what could go wrong.

01 Medical Ethics & Non-Diagnosis

! Risk: Users treating AI advice as clinical diagnosis → delayed treatment.

✓ Mitigation: Hard-coded Non-Doctor Mode. 48h+ persistent symptom forces 'Book a Doctor'. Referral PDF language is always 'Pointers for your doctor', never a diagnosis.

02 Global Privacy & Compliance

! Risk: Health data exposure across jurisdictions: DPDP, HIPAA, GDPR.

✓ Mitigation: Zero health data on servers. AES-256-GCM on device. sha256 user hash only (non-reversible). Compliant by architecture, not just policy.

03 AI Hallucination in Drug Safety

! Risk: Fabricated drug interactions directly harm patients on critical medications.

✓ Mitigation: Hard contraindication rules fire BEFORE Claude call. Claude instructed: never invent pharmacology. Severity='severe' always sets must_consult_doctor=True.

04 Rural Connectivity & Low-End Devices

! Risk: Core users: 2G networks, ₹3,000–₹6,000 Android phones, no data plan.

✓ Mitigation: Offline-first: trigger rules run locally. Cached STT models. WhatsApp bot fallback, no app install. Core bundle <5MB. Voice-only mode.

ANTICIPATED RESULTS & IMPACT PROJECTIONS

What Kutumb achieves — quantified, per persona, scalable.



PERSONA-WISE QUANTIFIED ADVANTAGE

R	Ramprasad (Elderly Parent)	Medicine adherence +70%. SOS alerts children in <60s. Doctor visits arrive with 6-month health data. Est. ₹18,000/year saved on avoidable consultations.	5 min/day voice check-in
P	Priya (Global Caregiver)	Real-time parent health dashboard from anywhere. 3h/week saved on remote care. 50% fewer ‘emergency flight’ scenarios. Peace of mind, quantified.	2 min/day dashboard review
S	Savitribai (ASHA Worker)	200 families managed from one screen. Wrong referrals cut by 50%. Maternal monitoring enabled. Fully offline-capable, works on 2G feature phone.	Weekly data sync in town

WHY KUTUMB: CREDIBILITY & GLOBAL VISION

Built in India. Designed for the world. Trusted by architecture.

WHY WE'RE CREDIBLE

A	Complete architecture built 12 FastAPI modules, 30+ endpoints, full Firestore schema, 3-layer Claude prompt. All documented, all buildable now.
B	Privacy-first by engineering AES-256-GCM local. Zero health data server storage. Meets DPDP, HIPAA, GDPR simultaneously — by architecture.
C	Claude AI as core brain Anthropic claude-sonnet-4-6 with hard ethical guardrails — no diagnosis, must_consult_doctor enforced, never invents pharmacology.
D	Evidence-based projections Every number grounded in WHO, ICMR, NLM, and peer-reviewed health economics. Not guesses — cited research.

GLOBAL EXPANSION ROADMAP

Phase 1 (Now) <i>India</i>	11 Indic languages, 300M+ families, ASHA + consumer app
Phase 2 (6mo) <i>SE Asia + E. Africa</i>	Swahili, Hausa, Tagalog, Bahasa. NGO + government partnerships
Phase 3 (18mo) <i>Latin America + MENA</i>	Spanish, Portuguese, Arabic. Telehealth integration. 3B speakers
Phase 4 (36mo+) <i>Global</i>	7,100 languages. White-label NHS, Medicaid, Ayushman Bharat

HopeWorks alignment: Kutumb reduces catastrophic health expenses — the single biggest driver of poverty in LMICs. Health equity = economic empowerment.

THANK YOU

for your time and attention

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Built for India. Designed for the world.

Team: Amrit & Nishita

WitchHunt 2026 · Health & Well-being · HopeWorks

Mission: Health for Every Human

India → SE Asia → Africa → Latin America → Global
7,100 languages. 1 platform. 1 goal.